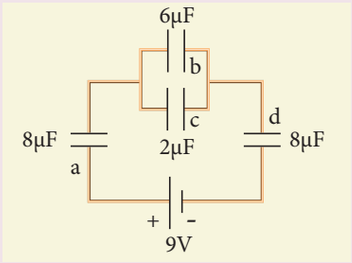
(a) If the bottom part of the cloud is 1000 m above the ground, determine the electric potential difference that exists between the cloud and ground.

(b) In a typical lightning phenomenon, around 25C of electrons are transferred from cloud to ground. How much electrostatic potential energy is transferred to the ground?

Ans: (a) $V = 3 \times 10^9$ V, (b) $U = 75 \times 10^9$ J

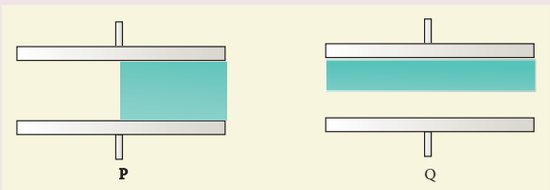
14. For the given capacitor configuration (a) Find the charges on each capacitor (b) potential difference across them (c) energy stored in each capacitor



Ans:

$Q_a = 24\ \mu C$, $Q_b = 18\ \mu C$,
$Q_c = 6\ \mu C$, $Q_d = 24\ \mu C$
$V_a = 3V$, $V_b = 3V$,
$V_c = 3V$, $V_d = 3V$,
$U_a = 36\ \mu J$, $U_b = 27\ \mu J$,
$U_c = 9\ \mu J$, $U_d = 36\ \mu J$

15. Capacitors P and Q have identical cross sectional areas A and separation d. The space between the capacitors is filled with a dielectric of dielectric constant ε_r as shown in the figure. Calculate the capacitance of capacitors P and Q.



Ans : $C_P = \frac{\varepsilon_o A}{2d}(1+\varepsilon_r)$, $C_Q = \frac{2\varepsilon_o A}{d}\left(\frac{\varepsilon_r}{1+\varepsilon_r}\right)$

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UNIT-1(XII-Physics_Vol-1).indd 80 04-03-2019 10:27:00